DS607 Group Project Proposal

Statistical Inference on a Rule-Based Bitcoin Trading Strategy

Course: DS607 Statistics for Data Science, AY2026/27 August Term (Instructor: Qiu Yue)
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Topic clearance: topic and data source approved by Prof Qiu on 17 September 2026

Inference question

Does a pre-specified 5-day / 20-day moving-average crossover rule applied to Bitcoin daily prices produce a mean daily excess return over buy-and-hold that is statistically distinguishable from zero?

Data source and series

- **Source:** Binance public REST API, market-data endpoint `GET /api/v3/klines`.
- **Series:** BTC/USDT spot, daily candles (UTC), closing prices.
- **Sample window:** 1 January 2019 to 31 December 2025 (2,557 daily returns). December 2018 is pulled only to initialise the moving averages.
- **Licensing:** the data are public and free to use. No account, API key, or SMU-licensed database is involved, so the raw data can be committed to the repository.

Methods

Let r_t be the daily log return. The signal $s_t = 1\{\text{SMA}_5 > \text{SMA}_{20}\}$ is computed from closes up to day $t-1$, so no future information enters the rule. The excess return is $d_t = s_t r_t - r_t$, and we test $H_0 \colon E[d_t] = 0$.

1. **Classical baseline:** one-sample t-test and CLT-based confidence interval for the mean of d_t. This treats the days as independent, and we use it as the independence-based baseline. We check that assumption directly (autocorrelation of d_t, volatility clustering, heavy tails).
2. **Moving block bootstrap:** confidence interval and test for the same mean, resampling blocks of consecutive days so that serial dependence is kept. Block length $\ell = \text{round}(n^{1/3}) = 14$, $B = 9{,}999$ resamples, fixed seed. We will document every implementation step (block drawing, block length, interval type, p-value) and report sensitivity to $\ell \in \{1, 5, 10, 20, 40\}$.
3. **Monte Carlo check:** simulated series with known mean zero (i.i.d., heavy-tailed, AR(1), GARCH) to compare the coverage of the two intervals.

All parameters (rule windows, sample, block length, sub-periods, cost) are fixed in advance and never tuned on the data. Robustness checks: a bull sub-period (2020–2021), a bear sub-period (2022), and a 0.1% cost per position switch. A permutation test is an optional extension.

Expected outputs

A GitHub repository following the course layout (README, AI_USAGE, CONTRIBUTIONS, requirements, notebooks, `outputs/`), with a side-by-side confidence-interval table for the two methods, a block-length sensitivity plot, a coverage table from the simulation, a cumulative-return plot, and a written discussion of limitations (single exchange, single rule, regime dependence, simplified costs).

Note on AI use: this proposal was drafted with the help of an AI writing tool (Claude) from the group's own design; all members reviewed it. Further AI use will be recorded in AI_USAGE.md.